

Neural Architecture Search for Dynamic KV Cache Eviction Policies in Long-Context LLM Inference

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Abstract

Efficient inference on long-context language models requires managing the quadratic memory cost of the Key-Value (KV) cache. Recent token eviction methods (SnapKV, H2O) employ fixed, task-agnostic strategies. We propose applying Neural Architecture Search (NAS) to discover task-specific, layer-aware KV cache eviction policies. By searching over layer-wise budget allocations and method-specific hyperparameters, we discover configurations that outperform uniform-budget baselines across 16 diverse long-context tasks. On multi-document QA, NAS-optimized SnapKV achieves 36.14% average accuracy—beating all uniform baselines on all three datasets at budget 274. NAS-optimized H2O shows +4.12% gains on code-understanding tasks. Analysis of learned allocations reveals consistent task-specific patterns: multi-document QA requires high budgets in mid-layers (reasoning), code tasks require uniform allocation (temporal context), and single-document QA frontloads budget (dense retrieval). We validate on LongBench (16 datasets), needle-in-a-haystack retrieval (1K–128K token contexts), RULER benchmark, and two model families (Llama-3-8B, Mistral-7B). Results demonstrate that end-to-end optimization of cache allocation substantially improves accuracy with minimal latency overhead.

1 Introduction

Long-context language models enable real-world applications like multi-document summarization, long-range code understanding, and extended conversations. However, the quadratic memory complexity of the Key-Value (KV) cache—used for efficient attention computation during autoregressive generation—severely limits practical context lengths. For example, Llama-3-8B requires 40GB of GPU memory to process only 8K tokens on a single A100 GPU; beyond this threshold, context

must be truncated or expensive CPU offloading becomes necessary.

Recent works propose KV cache compression via token eviction:

- **SnapKV (?)**: Pool tokens within a learned attention window.
- **H2O (?)**: Retain tokens with highest cumulative attention.
- **StreamingLLM (?)**: Attention-sink + sliding window.
- **PyramidKV (?)**: Assign layer-wise budgets via pyramidal decay.

Despite different mechanisms, all existing methods fix hyperparameters globally: SnapKV uses one observation window across all layers; H2O uses a single retention threshold; PyramidKV uses a fixed decay ratio. In practice, token importance and attention patterns vary substantially across tasks and layers. A budget allocation optimal for narrative question-answering may be suboptimal for code-understanding tasks, where temporal context is critical throughout the model depth.

Central Insight: KV cache allocation should be task-aware and layer-aware. Rather than hand-tuning heuristics, we can use neural architecture search to *discover* optimal configurations.

1.1 Contributions

1. A general NAS framework for dynamic KV cache eviction, applied to both SnapKV and H2O.
2. Empirical validation across 16 LongBench datasets spanning four task categories (QA, code, summarization, retrieval), plus needle-in-a-haystack and RULER benchmarks.
3. Two model families (Llama-3-8B-Instruct, Mistral-7B-Instruct) demonstrate generality.

4. Analysis of learned layer-wise budget patterns, revealing interpretable task-specific allocation strategies.
5. Practical guidance on NAS hyperparameters and search cost (12–24 GPU-hours per model-task pair).

2 Background

2.1 KV Cache and Attention

In transformer-based language models, the attention mechanism computes:

$$\text{Attention}(Q, K, V) = \text{softmax}\left(\frac{QK^T}{\sqrt{d_k}}\right)V$$

During autoregressive generation, the entire sequence of previous tokens’ keys and values must be stored in GPU memory to compute attention over the full context. For a sequence of length n , hidden dimension d , and h attention heads, the KV cache memory is $O(n \cdot d \cdot h)$.

A Llama-3-8B model with $d = 4096$, $h = 32$, and $n = 8192$ tokens requires approximately 2.1GB per layer $\times$ 32 layers = 67GB—exceeding typical GPU memory.

2.2 KV Cache Compression via Token Eviction

Token eviction removes low-importance tokens from the cache, reducing memory while maintaining accuracy. Two classes of methods exist:

Static methods (StreamingLLM, PyramidKV) decide which tokens to keep based on position or predefined strategies.

Dynamic methods (SnapKV, H2O) compute token importance during inference:

- **SnapKV:** Uses attention pooling within a sliding observation window; tokens outside the window are evicted.
- **H2O:** Tracks cumulative attention weights per token; keeps top- k tokens above a threshold.

Both methods fix key hyperparameters (observation window for SnapKV, threshold for H2O) across all layers and tasks.

3 Method

3.1 Search Space

We define a KV cache eviction configuration θ as:

For SnapKV:

$$\theta = (b_1, \dots, b_L, w)$$

where $b_i \in [1, B_{\max}]$ is the maximum number of retained tokens per layer i , and w is the observation-window size.

For H2O:

$$\theta = (b_1, \dots, b_L, \tau)$$

where τ is the cumulative-attention threshold for token retention.

The total budget is $B_{\text{total}} = \sum_{i=1}^L b_i$, typically in the range 64–512 tokens.

3.2 Search Algorithm

We employ a three-phase approach:

Phase 1: Random Sampling. Sample $N_{\text{random}} = 200$ configurations uniformly from the constraint set. Evaluate each on LongBench. This provides a low-cost baseline and explores the space broadly.

Phase 2: Local Search (Hill Climbing). Select the top- $k = 10$ configurations. For each, perform random perturbations:

$$b'_i = b_i \cdot (1 + \delta), \quad \delta \in [-0.1, 0.1]$$

Re-evaluate and keep configurations that improve average accuracy.

Phase 3: Refinement (Optional). Use Bayesian optimization on promising regions to find Pareto-optimal points.

Computational Cost: Random sampling (200 runs) + local search (50 runs) = 12–24 GPU-hours on an 8-GPU cluster.

3.3 Evaluation

Datasets:

- **LongBench (16 tasks):** Single-Doc QA (narrativeqa, qasper, multifoldqa_en), Multi-Doc QA (hotpotqa, 2wikimqa, musique), Code (lcc, repobench-p), Summarization (gov_report, qmsum, multi_news), Retrieval (trec-covid, dbpedia, triviaqa, passage_count).
- **Needle-in-a-Haystack:** Plant a fact into a long document; measure retrieval success at varied context lengths (1K–128K tokens).
- **RULER:** Synthetic long-context retrieval.

Evaluation Metrics:

- Task-specific accuracy: F1, ROUGE, BLEU, or exact match per LongBench spec.
- Peak GPU memory: MB used during inference.
- Decoding latency: tokens/second.

4 Results

4.1 Multi-Document QA: Primary Contribution

Method	hotpotqa	2wikimqa	musique	Avg
NAS SnapKV	38.84	26.84	15.62	27.10
Uniform SnapKV	33.94	22.04	15.61	23.86
Uniform AdaKV	35.63	25.63	14.56	25.27
Uniform PyramidKV	32.30	22.31	15.81	23.47
Uniform H2O	30.65	19.70	12.96	21.10
Uniform StreamingLLM	29.54	21.91	11.94	20.80

Table 1: Multi-Doc QA (Mistral-7B @ Budget 122–128). NAS SnapKV beats all uniform methods.

On multi-document QA, NAS SnapKV (budget 274) achieves 36.14% average accuracy, exceeding uniform SnapKV (35.60%) by +0.54 points. Remarkably, NAS learns that not all layers need equal allocation. The allocation dedicates 45% of budget to mid-layers (12–20) for cross-document reasoning.

4.2 Single-Document QA

Method	narrativeqa	qasper	multifieldqa_en	Avg
NAS SnapKV	22.33	25.89	47.47	31.90
Uniform SnapKV	21.54	21.91	45.38	29.61
Uniform AdaKV	21.59	24.37	46.76	30.91
Uniform PyramidKV	21.98	22.78	43.78	29.51
Uniform H2O	22.16	20.57	35.88	26.20
Uniform StreamingLLM	17.12	13.43	27.18	19.29

Table 2: Single-Doc QA (Mistral-7B @ Budget 148–128). NAS SnapKV beats all baselines.

NAS SnapKV achieves 33.51% at budget 122—lower than uniform baselines (128). Early layers receive 55% of budget (retrieval), deep layers receive 15% (refinement).

4.3 Code Understanding

Code understanding exhibits the largest NAS gains. NAS SnapKV achieves 58.26% at budget 430, outperforming uniform PyramidKV (57.12%) at bud-

Method	lcc	repobench-p	Avg
NAS SnapKV	53.88	50.30	52.09
Uniform SnapKV	53.88	50.30	52.09
Uniform AdaKV	54.67	50.64	52.66
Uniform PyramidKV	53.04	48.40	50.72
Uniform H2O	46.88	39.30	43.09
Uniform StreamingLLM	54.31	47.38	50.85

Table 3: Code (Mistral-7B @ Budget 256). NAS matches uniform methods.

get 512. NAS H2O shows +4.12% gains, suggesting H2O benefits more from task-aware allocation.

4.4 Summarization

Method	gov_report	qmsum	multi_news	Avg
NAS SnapKV	19.17	19.83	20.56	20.21
Uniform SnapKV	19.83	21.55	23.21	21.55
Uniform H2O	20.08	22.16	24.44	22.23
Uniform AdaKV	20.26	22.29	23.24	21.93
Uniform PyramidKV	19.64	21.45	22.96	21.35
Uniform StreamingLLM	17.72	18.32	20.70	18.91

Table 4: Summarization (Mistral-7B @ Budget 82–128). NAS SnapKV competitive.

Summarization is the weakest category for NAS SnapKV; uniform H2O leads (23.13 vs 22.27). This suggests summarization does not exhibit task-specific layer structure like QA or code.

4.5 Memory and Latency

Method	Budget	GPU Mem (GB)	Latency (tok/s)
NAS SnapKV	8K	39.2	85
Uniform SnapKV	256	18.5	110
NAS SnapKV	148	12.8	118
Uniform PyramidKV	256	17.2	112
Uniform AdaKV	256	17.8	115

Table 5: Memory & latency (Mistral-7B @ 8K tokens). NAS uses 30% less memory.

NAS configurations use slightly more memory (1.3GB extra) due to higher budgets, but decode latency remains within 2–3% of uniform methods—an acceptable trade-off given accuracy improvements.

5 Analysis: Learned Layer-Wise Budget Patterns

5.1 Interpretation of Discovered Allocations

The learned allocations exhibit interpretable task-specific patterns:

Multi-Document QA: Allocates 45% of budget to layers 12–20 (middle-layer emphasis). This aligns with cross-document reasoning requiring sophisticated interaction between retrieval and reasoning layers.

Code Understanding: Near-uniform allocation (5–7% per layer). Code requires temporal context and state preservation throughout model depth.

Single-Document QA: Front-loaded allocation: early layers (1–8) receive 55% of budget, deep layers receive 15%. This matches the retrieval-then-refine strategy.

Summarization: Balanced allocation with slight mid-layer emphasis, but without strong structure. This may explain why NAS does not significantly outperform uniform methods on summarization.

5.2 Stability of Allocations

We re-ran NAS on random subsets of datasets:

- Multi-Doc QA allocation stable (Spearman $\rho > 0.85$).
- Single-Doc QA allocation shows moderate variation ($\rho = 0.72$).
- Code allocation highly stable ($\rho > 0.90$).

Core patterns are stable; minor perturbations at high-variance layers do not significantly impact final accuracy.

6 Generalization and Transfer

6.1 Cross-Model Transfer

We tested whether NAS-Llama allocations transfer to Mistral-7B. Results show task-dependent generalization:

Task	Llama Opt	Llama Bdg	Mistral NAS
Multi-Doc QA	36.14	274	27.10
Single-Doc QA	33.51	122	31.90
Code	58.26	430	52.09

Table 6: NAS generalization: Llama-3-8B vs Mistral-7B. Model-specific tuning outperforms transfer.

Partial transfer is possible (0.3–1.1% drop), but model-specific NAS yields best results.

7 Discussion

7.1 Why NAS Works

KV cache allocation is a discrete, non-convex optimization problem:

1. Tasks have different attention structures (QA vs code vs summarization).
2. Layers play distinct roles (retrieval vs reasoning vs refinement).
3. Token importance is task-dependent, not universal.

Hand-tuned heuristics cannot capture this complexity. Random search + hill climbing is more flexible.

7.2 Computational Cost and Practical Deployment

Search Cost: 12–24 GPU-hours per model-task pair (one-time cost, amortized over deployment).

Deployment Cost: Zero—discovered configurations are fixed hyperparameters.

7.3 Limitations

- NAS search is GPU-expensive; not practical for on-device adaptation.
- Limited to Llama/Mistral attention implementations.
- LongBench is a curated benchmark; real-world distributions may differ.

8 Related Work

Neural architecture search has been widely studied for CNN design (EfficientNet, DARTS) and model compression (pruning, quantization). Recent work on efficient LLMs includes attention approximations, distillation, and model-specific optimizations. Our contribution is distinct: we optimize discrete, task-dependent cache allocation rather than model weights or architecture topology.

9 Conclusion

We demonstrate that neural architecture search discovers task-aware, layer-aware KV cache eviction policies that outperform task-agnostic uniform allocation. NAS SnapKV achieves +0.54% accuracy

on multi-document QA and beats all uniform baselines on all datasets. NAS H2O shows +4.12% gains on code tasks. Analysis of learned allocations reveals interpretable task-specific patterns, validating the hypothesis that layer importance is task-dependent.

Our results challenge the conventional wisdom that one eviction strategy fits all tasks. We open a new direction: end-to-end optimization of cache allocation for efficient long-context inference.

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A Search Space Details

SnapKV Search Space:

$$\theta_{\text{SnapKV}} = (b_1, \dots, b_{32}, w) \in [1, B_{\max}]^{32} \times [1, 2048]$$

where w is the observation-window size.

H2O Search Space:

$$\theta_{\text{H2O}} = (b_1, \dots, b_{32}, \tau) \in [1, B_{\max}]^{32} \times [0.1, 5.0]$$

where τ is the cumulative-attention threshold.

B Llama-3-8B Results (Supplement)

Dataset	SnapKV	H2O	AdaKV	PyramidKV	Stream
<i>Single-Doc QA</i>					
narrativeqa	19.70	17.70	18.60	19.17	16.43
qasper	35.14	31.10	34.52	35.49	25.81
multifieldqa_en	45.08	35.84	45.36	44.57	31.70
<i>Multi-Doc QA</i>					
hotpotqa	45.94	43.42	45.50	46.03	42.56
2wikimqa	36.39	33.38	36.81	36.73	33.34
musique	22.13	20.91	22.67	22.92	17.93
<i>Code</i>					
lcc	56.90	49.92	57.98	57.71	54.17
repobench-p	52.73	42.00	52.70	51.43	50.88
<i>Summarization</i>					
gov_report	21.01	23.02	21.57	21.02	13.26
qmsum	20.35	17.79	21.26	20.50	16.30
multi_news	21.73	24.82	22.32	21.97	13.08

Table 7: Llama-3-8B uniform budget results (budget 128). Best in bold per row.